

Origins Data Science Lab

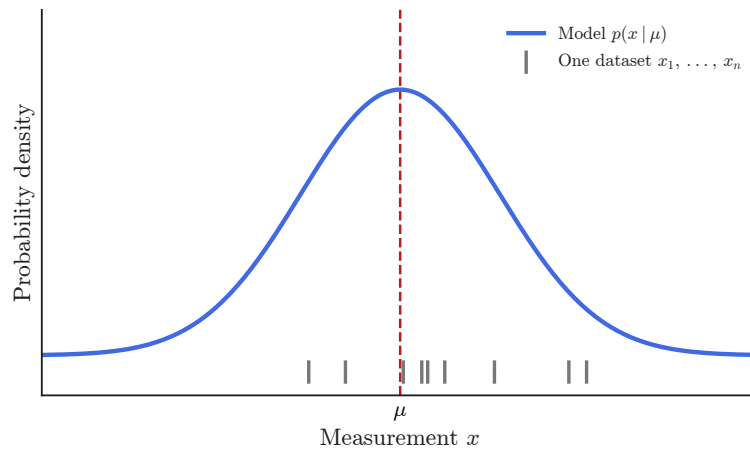
Bayesian & Frequentist Probability

Exam Guidance Questions

These questions outline the **provisional scope** of the 90-minute exam. You may bring one self-made A4 sheet, handwritten on both sides (tablet handwriting allowed). Use the hints to develop short derivations and clear interpretations, stating your assumptions.

1. Sampling Distributions, Likelihood, and Estimation

- For independent $\mathcal{X} = \{x_1, \dots, x_n\}$ drawn from $\mathcal{G}[x \mid \mu, \sigma^2]$ with known σ^2 , construct the likelihood $\mathcal{L}(\mu)$ and derive $\hat{\mu}_{\text{MLE}}$.
- Derive $\mathbb{E}_\mu[\hat{\mu}]$, $\text{Var}_\mu(\hat{\mu})$ and $p(\hat{\mu} \mid \mu)$. Is the Gaussian sampling distribution exact or a CLT approximation?



Gaussian model for one measurement with a sampled dataset; the dashed line marks μ .

2. Fisher Information and the Cramér–Rao Bound

- Define the score $s(\pi)$ and Fisher information $F[\pi]$. Calculate $F[\mu]$ for the full dataset in Question 1. How does it scale with n ?
Hint: Start from $s(\pi) = \nabla_\pi \log \mathcal{L}$; the expectation defining Fisher information averages over data at fixed π .
- State the Cramér–Rao bound (for an unbiased estimator) and its assumptions. Does $\hat{\mu} = \bar{x}$ attain it, and what does this say about its efficiency (how close its variance is to the Cramér–Rao lower bound)?
- For $\hat{x} \sim \mathcal{G}[\hat{x} \mid x[\pi], \sigma^2]$, with scalar $\pi \in \mathbb{R}$ and known, fixed variance σ^2 , how does linear versus nonlinear $x[\pi]$ affect the shape of the likelihood in π ? In the linear case, assume a nonzero slope.
Hint: Consider how π enters $(\hat{x} - x[\pi])^2$ in the log-likelihood.

3. Bias, Variance, and Nonlinear Transformations

- For scalar π , define $\text{Bias}_\pi[\hat{\pi}]$, $\text{Var}_\pi[\hat{\pi}]$ and $\text{MSE}_\pi[\hat{\pi}]$. If $\hat{\pi}$ is unbiased, compare it with $\hat{\pi}_\lambda = \lambda \hat{\pi}$ for fixed $0 < \lambda < 1$. When can this reduce MSE?

Hint: Use $\mathbb{E}[aY] = a\mathbb{E}[Y]$, $\text{Var}(aY) = a^2 \text{Var}(Y)$ and $\text{MSE} = \text{Var} + \text{Bias}^2$.

- For unbiased $\hat{\pi}$, explain why $g(\hat{\pi})$ can be biased for $g(\pi)$. Obtain a second-order bias approximation and state when it is useful.

Hint: Expand about π , then use $\mathbb{E}_\pi[\hat{\pi} - \pi] = 0$ for an unbiased estimator.

- Find the exact bias for $g(\hat{\pi}) = \hat{\pi}^2$.

Hint: Express $\mathbb{E}[\hat{\pi}^2]$ using its variance and mean.

4. Gaussian Priors and Posteriors

- For a scalar mean μ , use the Gaussian prior $p(\mu) = \mathcal{G}[\mu \mid \mu_p, \sigma_p^2]$ and sampling distribution $p(\hat{\mu} \mid \mu) = \mathcal{G}[\hat{\mu} \mid \mu, \sigma^2/n]$. Derive the posterior mean μ_{post} and variance σ_{post}^2 .

Hint: Use $p(\mu \mid \hat{\mu}) \propto p(\hat{\mu} \mid \mu)p(\mu)$. Collect powers of μ in $n(\bar{x} - \mu)^2/\sigma^2 + (\mu - \mu_p)^2/\sigma_p^2$ and complete the square.

For the remaining questions, you may use the following scalar results even if you have not derived them:

$$\mu_{\text{post}} = \frac{n\bar{x}/\sigma^2 + \mu_p/\sigma_p^2}{n/\sigma^2 + 1/\sigma_p^2}, \quad \sigma_{\text{post}} = \left(\frac{n}{\sigma^2} + \frac{1}{\sigma_p^2} \right)^{-1/2}.$$

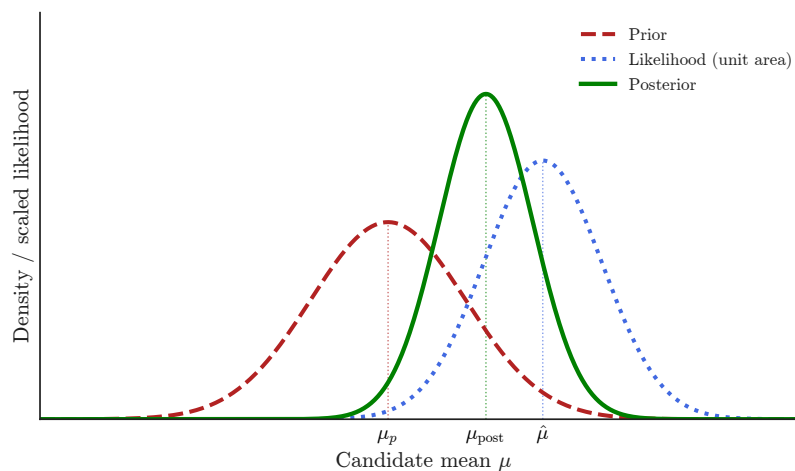
Here σ_{post} is the posterior *standard deviation*; its variance is σ_{post}^2 . The posterior is $p(\mu \mid \hat{\mu}) = \mathcal{G}[\mu \mid \mu_{\text{post}}, \sigma_{\text{post}}^2]$.

- Explain how the prior and data determine the posterior mean. How does the prior pull the posterior mean towards μ_p , and what happens for a weak prior?

Hint: Precision means inverse variance. Compare $1/\sigma_p^2$ with n/σ^2 ; take $\sigma_p^2 \rightarrow \infty$ with μ_p fixed.

- Distinguish the posterior variance σ_{post}^2 from the sampling variance of $\hat{\mu}_{\text{MAP}}$.

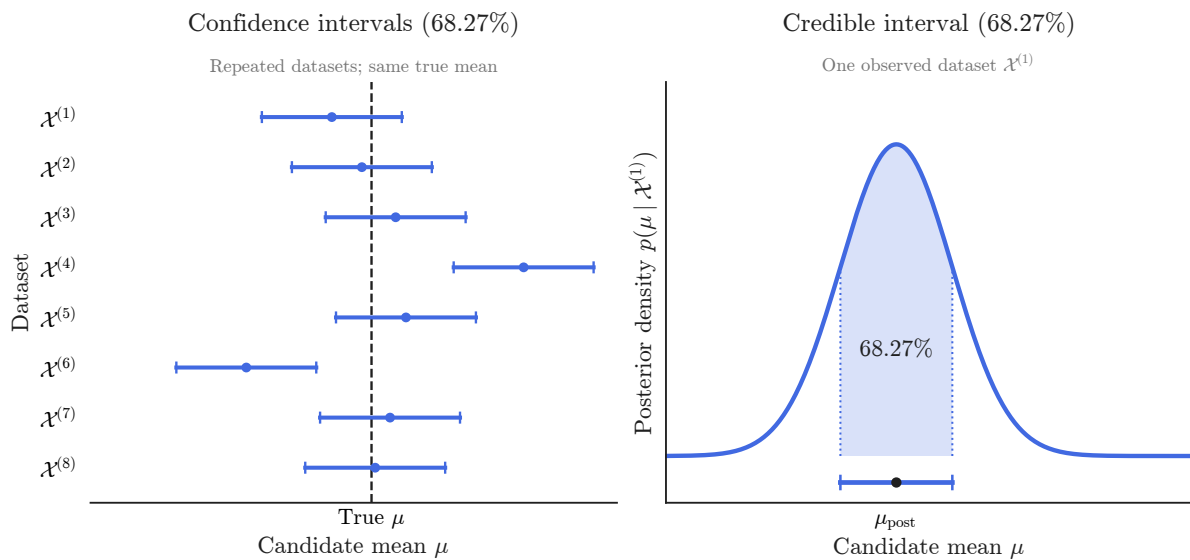
Hint: For this Gaussian posterior, $\hat{\mu}_{\text{MAP}} = \mu_{\text{post}}$. Contrast varying μ within one posterior with recomputing its mean from many datasets generated at a fixed true μ .



Prior and posterior densities, with the likelihood rescaled to unit area for comparison.

5. Confidence, Credibility, and Coverage

- In one or two sentences, explain what probability means to a frequentist and to a Bayesian. How does each approach treat an unknown parameter?
- Construct a central 68.27% confidence interval $\mathcal{C}_{68.27\%}^{\text{CI}}$ for a scalar Gaussian mean with known σ^2 . Explain which quantity is fixed and which varies in its coverage statement.
Hint: Standardise $\hat{\mu} - \mu$ by the standard deviation of $\hat{\mu}$. A standard Gaussian lies between -1 and 1 with probability approximately 0.6827; rearrange to obtain a set of μ .
- Construct the corresponding Gaussian credible interval $\mathcal{C}_{68.27\%}^{\text{CR}}$ and explain its probability statement. When can its endpoints agree with those of the confidence interval?
Hint: Use μ_{post} and σ_{post} ; integrate the posterior over the interval. Examine the weak-prior limit.
- If we repeat the experiment many times at a fixed true parameter, must 68.27% credible intervals contain that value about 68.27% of the time? Briefly explain your answer.



Confidence intervals across repeated datasets (left), and a credible interval for the first dataset (right).